Analytic interdomain memory for efficient online HiPPO-SVGP

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Abstract

Online Gaussian processes (GPs) are attractive for streaming prediction because they combine flexible function modeling with calibrated uncertainty, but exact inference is cubic and standard sparse online updates can forget early observations. Online HiPPO Sparse Variational Gaussian Processes (OHSVGP) addresses this issue by using HiPPO-based interdomain inducing variables, yet its interdomain kernel construction is built from temporal ODE recursion of Fourier–Legendre states. We study an analytic replacement for this kernel-construction step in the one-dimensional HiPPO-LegS setting with stationary kernels. The main idea is to express the required sine/cosine–Legendre overlap integrals in closed form through spherical Bessel functions, so that kernel quantities are evaluated directly at time t and the remaining recurrence is over polynomial order rather than over timestamps. This changes the recursion structure of interdomain kernel construction, removes ODE discretization error from that construction, and reduces its asymptotic dependence on the number of ODE integration steps needed to reach the current horizon. On Solar Irradiance and COVID-19 mortality streams, the resulting analytic variant maintains comparable predictive performance and is slightly favorable on several aggregate metrics relative to ODE-based OHSVGP, and runtime scaling on the solar benchmark confirms the predicted reduction in discretization-step dependence, with the strongest gains at fine time discretizations.

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1. Introduction

Gaussian processes (GPs) remain a standard probabilistic tool for time-series modeling since they combine nonparametric flexibility with principled uncertainty quantification (Roberts et al., 2013). Their deployment in online settings, however, is limited by the cubic complexity of exact inference and by the fact that sparse online approximations (Titsias, 2009; Hensman et al., 2013; Bui et al., 2017; Maddox et al., 2021) often struggle with catastrophic forgetting.

The Online HiPPO Sparse Variational Gaussian Process (OHSVGP) framework (Chen et al., 2025) addresses this issue by constructing the inducing variables for GPs with high-order polynomial projection operators (HiPPO) (Gu et al., 2020), substantially alleviating catastrophic forgetting in online regression and continual-learning settings. By defining inducing features as HiPPO's time-varying orthogonal polynomials over $[0, t]$, a fixed set of inducing variables optimally compresses the entire observation history as the horizon grows.

Despite these benefits, the original implementation relies on ODE-based evolution of the interdomain features, which requires discretization to integrate over time. Therefore, as finer ODE discretizations are used to reach the current horizon, integration becomes computationally expensive and discretization errors accumulate. We observe that for HiPPO-LegS in one dimension, the sine/cosine–Legendre inner products required by the interdomain construction admit closed forms via spherical Bessel functions, removing the need for ODE recursion altogether.

Our contributions are threefold. First, we show that the sine/cosine–Legendre overlaps underlying the OHSVGP interdomain random-feature matrix admit closed forms via spherical Bessel functions, yielding a direct analytic construction of $Z(t)$ and hence the same RFF approximations to $K_{fu}(t)$ and $K_{uu}(t)$ without temporal ODE evolution. Second, we demonstrate that the analytic construction avoids per-step ODE recursion, reducing kernel cost significantly when the number of ODE integration steps far exceeds the basis size, and eliminates the discretization error that accumulates over long horizons. Third, we validate on Solar Irradiance and COVID-19 mortality streams that the analytic variant preserves predictive behavior comparable to ODE-based OHSVGP.

2. Background

Here we review the necessary background on GPs and their sparse approximations, and interdomain GPs, then introduce the online HiPPO sparse GP framework which motivates our analytic extension.

2.1. Gaussian Processes, Sparse Variational Inference, and Online Updates

Let $f \sim \mathcal{GP}(0, k)$ be a Gaussian process with covariance kernel k (Rasmussen and Williams, 2006). Given data $\{(x_i, y_i)\}_{i=1}^N$, exact GP inference scales cubically in N,

motivating sparse variational approximations based on inducing variables u (Titsias, 2009; Hensman et al., 2013). In the standard formulation, the variational posterior is defined as

$$q(f, u) = p(f \mid u)q(u), \tag{1}$$

where $q(u)$ is usually Gaussian with free mean and covariance. For Gaussian likelihoods, collapsed objectives are available; for non-conjugate likelihoods, stochastic variational inference is typically used (Hensman et al., 2015).

Interdomain GPs generalize inducing variables from point evaluations to integral transforms,

$$u_m = \int f(x)\phi_m(x)\,dx, \tag{2}$$

where $\{\phi_m\}$ are basis functions chosen to induce desirable structure (Lázaro-Gredilla and Figueiras-Vidal, 2009; van der Wilk et al., 2020). The corresponding covariances are

$$[K_{fu}]_m = \int k(x, x')\phi_m(x')\,dx', \tag{3}$$

$$[K_{uu}]_{nm} = \iint k(x, x')\phi_n(x)\phi_m(x')\,dx\,dx'. \tag{4}$$

In online learning, data arrive sequentially. Standard online sparse GP and online sparse variational GP methods update the posterior using only new data and the previous variational approximation (Csató and Opper, 2002; Bui et al., 2017; Maddox et al., 2021). While computationally attractive, this approach does not guarantee memory retention, as inducing locations remain optimized relative to recent observations.

At update step t, such methods optimize a variational objective on the newly arrived batch while regularizing the new variational posterior toward the information carried forward from the previous step:

$$\mathcal{L}_t(q_t) = \mathbb{E}_{q_t(f)}[\log p(y_t \mid f)] - \mathrm{KL}[q_t(u) \,\|\, \widetilde{p}_{t-1}(u)]\,, \tag{5}$$

where $\widetilde{p}_{t-1}(u)$ denotes the predictive prior or transferred variational information from the previous update. In this paper, the online variational objective is unchanged from OHSVGP; the 'one ELBO update step' runtime in Section 4.3 refers to evaluating this objective, backpropagating through it, and taking one optimizer step.

2.2. HiPPO, Recurrent Memory, and OHSVGP

HiPPO constructs a finite-dimensional memory state by projecting historical signals onto time-varying orthogonal polynomial bases (Gu et al., 2020). For HiPPO-LegS, the basis functions are scaled Legendre polynomials over the interval $[0, t]$, producing coefficients that satisfy a linear ODE:

$$\frac{d}{dt}c(t) = A(t)c(t) + B(t)f(t), \tag{6}$$

supporting streaming updates and long-range memory.

Building on this memory construction, OHSVGP turns HiPPO coefficients into interdomain inducing variables (Chen et al., 2025),

$$u_m(t) = \int f(x)\phi_m^{(t)}(x)\,dx, \tag{7}$$

with time-dependent basis $\phi_m^{(t)}$. In the one-dimensional HiPPO-LegS case considered here, the basis functions on $[0, t]$ are

$$\phi_\ell^{(t)}(x) = \frac{2\ell+1}{t} P_\ell\left(\frac{2x}{t} - 1\right)\mathbf{1}_{[0,t]}(x), \tag{8}$$

We use this normalization throughout the paper; alternative orthonormal Legendre conventions differ only by row-wise scaling factors in the feature matrix. These basis functions induce the interdomain quantities

$$[K_{fu}(t)]_{n\ell} = \int_0^t k(x_n, x)\phi_\ell^{(t)}(x)\,dx, \tag{9}$$

$$[K_{uu}(t)]_{\ell m} = \int_0^t \int_0^t k(x, x')\phi_\ell^{(t)}(x)\phi_m^{(t)}(x')\,dx\,dx'. \tag{10}$$

In the original OHSVGP implementation, these kernel terms are built in Random Fourier Feature (RFF) space (Rahimi and Recht, 2007; Hensman et al., 2018) by evolving Fourier-Legendre coefficients recursively in t with a HiPPO ODE, so that kernel construction is tied to a time-stepping procedure whose cost and discretization error both grow with the number of ODE integration steps.

2.3. Spectral Factorization of the Interdomain Kernels

Starting from the HiPPO-LegS basis and interdomain kernels introduced above, we next recall the spectral factorization that underlies the analytic construction. We assume throughout that the kernel k is stationary and admits a spectral density $p(\omega)$, so that Bochner's theorem gives (Rudin,

1990; Rahimi and Recht, 2007)

$$k(x,x') = \mathbb{E}_{\omega\sim p(\omega)}\big[\cos(\omega x)\cos(\omega x') + \sin(\omega x)\sin(\omega x')\big]. \quad (11)$$

Drawing R independent frequencies $\{\omega_r\}_{r=1}^{R} \sim p(\omega)$ replaces the expectation by an empirical average. Substituting this factorization into $K_{fu}(t)$ and $K_{uu}(t)$ and exchanging expectation with integration over x shows that each sampled frequency ω contributes to the kernels only through a pair of one-dimensional projections of the HiPPO-LegS basis:

$$C_\ell(t;\omega) = \int_0^t \cos(\omega x)\,\phi_\ell^{(t)}(x)\,dx, \quad (12)$$

$$S_\ell(t;\omega) = \int_0^t \sin(\omega x)\,\phi_\ell^{(t)}(x)\,dx. \quad (13)$$

Stacking the cosine and sine projections over all sampled frequencies gives the interdomain random-feature matrix $Z(t) \in \mathbb{R}^{M\times 2R}$. For column index $q = 1,\dots,2R$, define

$$Z(t)_{\ell,q} = R^{-1/2}\begin{cases} C_\ell(t;\omega_q), & 1 \le q \le R, \\ S_\ell(t;\omega_{q-R}), & R < q \le 2R. \end{cases} \quad (14)$$

The corresponding data-side RFF matrix $Z_f(X) \in \mathbb{R}^{N\times 2R}$ is

$$Z_f(X)_{n,q} = R^{-1/2}\begin{cases} \cos(\omega_q x_n), & 1 \le q \le R, \\ \sin(\omega_{q-R} x_n), & R < q \le 2R. \end{cases} \quad (15)$$

The RFF approximation then assembles the interdomain covariance matrices as

$$K_{uu}(t) \approx Z(t)Z(t)^\top, \quad (16)$$

$$K_{fu}(t) \approx Z_f(X)Z(t)^\top. \quad (17)$$

Thus, after the Bochner/RFF reduction, the computational bottleneck is not the matrix product itself but the construction of $Z(t)$, i.e., the evaluation of the $2MR$ sine/cosine–Legendre overlaps at the current horizon t. The original OHSVGP construction obtains these overlaps by evolving Fourier-feature inputs through the HiPPO-LegS ODE in time (Chen et al., 2025), whereas our contribution is to compute the same overlaps directly in closed form.

3. Methods

We now turn to the new part of the construction. In the one-dimensional HiPPO-LegS setting (Gu et al., 2020; Chen et al., 2025) with a stationary kernel, the overlap integrals identified in Section 2.3 can be evaluated at a target time t in closed form, without the temporal HiPPO ODE recursion that the baseline OHSVGP uses to construct them. The derivation uses the plane-wave/Legendre identity and spherical Bessel functions (DLMF) to express every entry of the RFF feature matrix $Z(t)$ underlying $K_{fu}(t)$ and $K_{uu}(t)$, after which the sequence $\{j_\ell\}$ is generated by a short recursion over polynomial order rather than over time.

We therefore present the method in two steps. Section 3.1 derives the closed-form expression of each overlap via the plane-wave/Legendre identity. Section 3.2 describes the polynomial-order recursion used to evaluate the resulting spherical Bessel sequence and summarizes the computational cost. Figure 1 contrasts the full construction with the baseline ODE-based assembly.

3.1. Closed-Form via Spherical Bessel Functions

The key observation is that a change of variables to $u = 2x/t - 1$ turns C_ℓ and S_ℓ into the classical plane-wave/Legendre integral, which is known in closed form via spherical Bessel functions. The choice of u matches the argument of the Legendre polynomial in $\phi_\ell^{(t)}$, which already has the form $P_\ell(2x/t-1)$, and maps $[0,t]$ to the standard Legendre domain $[-1,1]$.

We combine C_ℓ and S_ℓ into a single complex integral

$$C_\ell(t;\omega) + i\,S_\ell(t;\omega) = \int_0^t e^{i\omega x}\,\phi_\ell^{(t)}(x)\,dx, \quad (18)$$

and introduce

$$u = \frac{2x}{t} - 1, \qquad \kappa = \frac{\omega t}{2}. \quad (19)$$

After substitution, the Legendre polynomial depends on u, and the exponential becomes a plane wave $e^{i\omega x} = e^{i\kappa(u+1)} = e^{i\kappa}\,e^{i\kappa u}$. Applying the classical plane-wave/Legendre identity (DLMF)

$$\int_{-1}^{1} e^{i\kappa u}\,P_\ell(u)\,du = 2\,i^\ell\,j_\ell(\kappa), \quad (20)$$

where j_ℓ is the spherical Bessel function of the first kind, closes the remaining integral. Factoring out $e^{i\kappa}$ and using $i^\ell = \cos(\ell\pi/2) + i\sin(\ell\pi/2)$, the real and imaginary parts give the closed-form overlaps

$$C_\ell(t;\omega) = (2\ell+1)\,j_\ell(\kappa)\,\cos(\kappa + \ell\pi/2), \quad (21)$$

$$S_\ell(t;\omega) = (2\ell+1)\,j_\ell(\kappa)\,\sin(\kappa + \ell\pi/2), \quad (22)$$

with $\kappa = \omega t/2$.

In these closed forms, t enters only through the scalar $\kappa = \omega t/2$; the polynomial order ℓ enters through the Bessel index and the phase shift. Using the RFF assembly introduced in Section 2.3, these closed-form overlaps define the same interdomain feature matrix $Z(t)$ used by the ODE-based OHSVGP construction. The difference is therefore not the

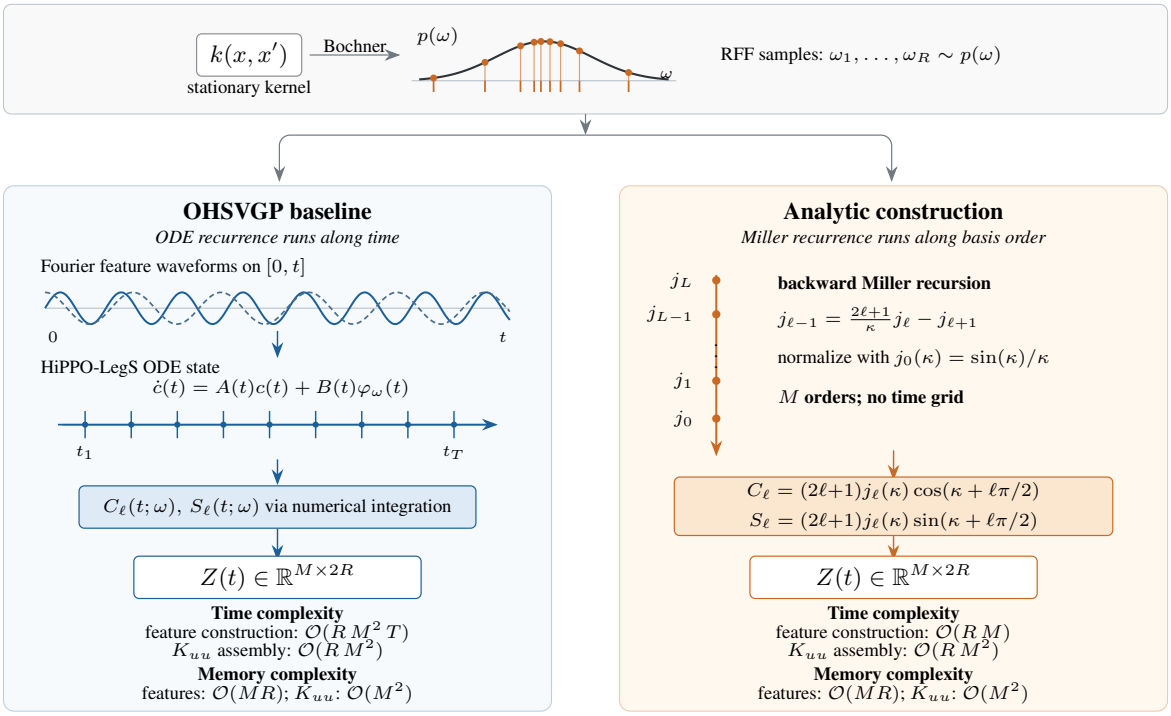


Figure 1. Two ways to assemble the HiPPO-LegS interdomain features $Z(t)$ in RFF space. Both start from the Bochner sampling of R frequencies $\{\omega_r\} \sim p(\omega)$. OHSVGP (left, blue) evaluates the sine/cosine–Legendre inner products by feeding the Fourier feature waveforms through an M-dimensional HiPPO-LegS ODE whose state is advanced at every discretization step, so the recursion runs along time with time complexity $\mathcal{O}(RM^2T)$. Our analytic construction (right, orange) instead runs a backward Miller recursion along the basis order ℓ (starting from a seed at $\ell = L \gg M$ and descending to $\ell = 0$), which returns all spherical Bessel values in $\mathcal{O}(M)$ per frequency; the projections C_ℓ, S_ℓ then follow in closed form. The axis of recursion therefore switches from time t to basis order ℓ, eliminating both the T factor and the accumulated discretization error. Both constructions produce the same feature matrix $Z(t) \in \mathbb{R}^{M\times 2R}$ and have the same leading feature-storage cost $\mathcal{O}(MR)$, with an additional $\mathcal{O}(M^2)$ storage for the inducing covariance when $K_{uu}(t)$ is materialized; the difference is in the temporal computation.

variational objective or the RFF kernel factorization, but the way $Z(t)$ is obtained: directly at the target horizon t, rather than by advancing a HiPPO-LegS ODE over all preceding discretization steps.

3.2. Numerical Evaluation and Complexity

The only remaining computational task is the evaluation of $\{j_\ell(\kappa)\}_{\ell=0}^{M-1}$ for each sampled frequency. The three-term recurrence

$$j_{\ell-1}(\kappa) = \frac{2\ell+1}{\kappa}\, j_\ell(\kappa) - j_{\ell+1}(\kappa) \tag{23}$$

is numerically unstable when applied upward from j_0, j_1 once ℓ exceeds κ, because it amplifies the irregular Bessel solution. We therefore evaluate it by backward Miller recursion (Gautschi, 1967): starting from a sufficiently large order $L \gg M$ with an arbitrary non-zero seed pair (j_L, j_{L+1}), we apply the recurrence downward to $\ell = 0$ and rescale the whole sequence so that $j_0(\kappa) = \sin(\kappa)/\kappa$ matches its known value. For small κ, Taylor expansions around the origin avoid division-by-zero in this normalization. The procedure is purely arithmetic and compatible with auto-differentiation, and it yields all M orders in $\mathcal{O}(M)$ work per frequency.

Figure 2 reports a representative convergence check for the

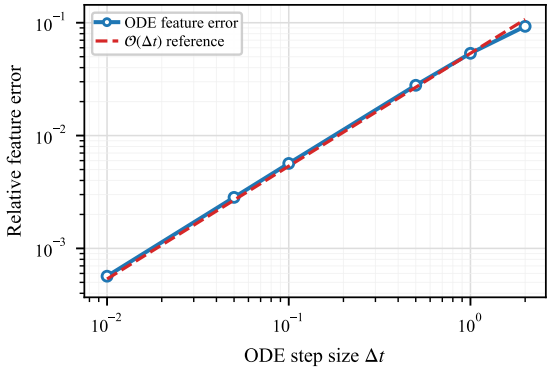


Figure 2. Convergence check for the ODE-based feature construction against the analytic evaluation. The horizontal axis is the ODE step size Δt; smaller values correspond to finer time discretization. The relative feature error follows the $\mathcal{O}(\Delta t)$ reference trend.

time-stepped ODE construction against the analytic feature evaluation. We measure

$$\mathrm{RelErr}(\Delta t) = \frac{\|Z_{\mathrm{ODE},\Delta t}(t) - Z_{\mathrm{analytic}}(t)\|_F}{\|Z_{\mathrm{analytic}}(t)\|_F}. \quad (24)$$

The relative feature error decreases as the ODE step size Δt becomes smaller and follows an approximately first-order trend over the tested range. This diagnostic is intended to verify the ODE feature discretization rather than to compare predictive performance on a particular dataset. It supports the interpretation that the analytic construction removes ODE time-stepping error from feature evaluation itself, while the remaining approximation error is due to the finite spectral representation rather than temporal discretization.

Next, we compare the time and memory complexity of the ODE-based and analytic constructions.

For a stream discretized into T ODE integration steps, the ODE-based construction need not store the full trajectory: it maintains only the current Fourier–Legendre projection state for each sampled frequency, requiring $\mathcal{O}(RM)$ feature-state memory and $\mathcal{O}(M^2)$ additional storage for $K_{uu}(t)$. Its time cost, however, remains sequential in the number of integration steps. Computing $Z(t)$ at the final horizon requires integrating the HiPPO-LegS feature ODE over $[0, t]$, equivalently applying T recurrent updates; with dense M-dimensional transitions, this costs $\mathcal{O}(RM^2T)$. In contrast, the analytic construction evaluates $Z(t)$ directly at the target horizon. The Bessel-based projections cost $\mathcal{O}(RM)$, while the dominant inducing-kernel assembly $K_{uu}(t) \approx Z(t)Z(t)^\top$ costs $\mathcal{O}(RM^2)$; constructing $K_{fu}(t) \approx Z_f(X)Z(t)^\top$ costs $\mathcal{O}(NRM)$ for N data inputs. Thus, the analytic construction removes the linear dependence on the number of preceding ODE integration steps while retaining the same leading storage order. Section 4.3 verifies this reduced discretization-step dependence in wall-clock measurements and shows that, at sufficiently fine temporal discretization, the analytic construction also yields faster kernel construction across the tested basis sizes.

4. Experiments

We evaluate the analytic construction against the ODE baseline on two streaming benchmarks: solar irradiance with a Gaussian likelihood and COVID-19 mortality with a Negative Binomial likelihood. Our goal is not to establish a new state-of-the-art among all online GP methods, but to isolate the effect of replacing the ODE-based interdomain feature construction inside OHSVGP.

4.1. Solar Irradiance

We evaluate a matched analytic-versus-ODE ablation of the OHSVGP framework on 10 sequential solar irradiance tasks with a Gaussian likelihood. The two variants share the same data split, kernel family, initialization, inducing budget, and spectral budget. We report task-wise RMSE and predictive negative log-likelihood (NLL), and defer representative posterior plots to Appendix D.

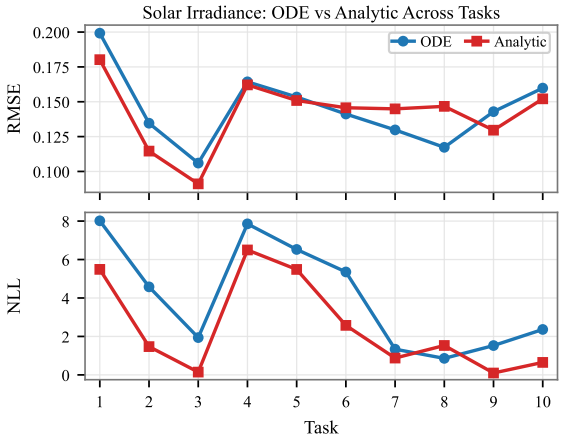


Figure 3. Task-wise RMSE and NLL across the 10-task solar stream.

Table 1. Solar irradiance comparison across 10 sequential tasks. Values are mean ± standard deviation across tasks, not random seeds. Lower is better. The main Solar runs use basis size $M = 100$ and ODE step size $\Delta t = 2.45$. Full task-wise values are given in Appendix D.

Method	RMSE ↓	NLL ↓	RMSE wins ↑	NLL wins ↑
ODE	0.145 ± 0.025	4.035 ± 2.637	3/10	1/10
Analytic	$\mathbf{0.142 \pm 0.024}$	$\mathbf{2.482 \pm 2.307}$	**7/10**	**9/10**

Table 1 summarizes the solar ablation over the 10-task stream. On this benchmark, the analytic construction preserves predictive behavior and improves aggregate RMSE and NLL, reducing mean RMSE from 0.145 to 0.142 and mean NLL from 4.035 to 2.482. The analytic model achieves lower RMSE on 7/10 tasks and lower NLL on 9/10 tasks, with the largest gain on Task 2 and a clear counterexample on Task 8.

Figure 3 visualizes the stream-level metric trends, while Appendix D gives representative posterior comparisons. Tasks 1 and 10 illustrate that the analytic variant remains competitive in both early and late stages of the stream, while Task 2 highlights the largest gain in both RMSE and NLL. We also include Task 8 as a failure case, where the ODE variant outperforms the analytic approximation.

4.2. COVID-19 Mortality

We next evaluate the same matched analytic-versus-ODE comparison on 5 sequential COVID-19 mortality tasks with a Negative Binomial likelihood. Here we summarize all 15 seen-task evaluation pairs induced by the 5-task stream. Our primary metrics are negative log predictive density (NLPD)

and expected calibration error (ECE), while RMSE and R^2 are reported as secondary diagnostics in Appendix E.

Table 2. COVID-19 mortality comparison across all 15 seen-task evaluation pairs. Values are mean ± standard deviation across evaluation pairs, not random seeds. Lower is better. The main COVID runs use basis size $M = 15$ and ODE step size $\Delta t = 2$. Full task-wise values are given in Appendix E.

Method	NLPD ↓	ECE ↓	NLPD wins ↑	ECE wins ↑
ODE	3.548 ± 0.410	0.109 ± 0.041	4/15	8/15
Analytic	$\mathbf{3.539 \pm 0.430}$	$\mathbf{0.109 \pm 0.038}$	**11/15**	7/15

Table 2 reports the COVID comparison. Averaged over all 15 seen-task evaluation pairs, the analytic variant yields slightly lower mean NLPD (3.539 vs. 3.548) and essentially matching mean ECE (both ≈ 0.109). It achieves lower NLPD on 11/15 pairs and lower ECE on 7/15 pairs, indicating modest aggregate gains in probabilistic accuracy. The final stage after Task 5 is more mixed: the analytic model still attains slightly lower mean NLPD (3.714 vs. 3.719) and slightly higher mean R^2 (0.701 vs. 0.699), but its mean ECE worsens from 0.101 to 0.114 because of a clear regression on the final held-out Task 5 evaluation. These results support predictive parity with the ODE construction rather than a uniform accuracy improvement.

These aggregate results are not uniform across the stream. The analytic variant remains close to the ODE baseline in RMSE and R^2 on most evaluation pairs, and its probabilistic gains are concentrated in a subset of mid-stream evaluations rather than being distributed evenly. The clearest counterexample is the final Eval 5 after Task 5 pair, where calibration degrades enough to offset some of the earlier ECE improvements. This pattern is consistent with a method that largely preserves the baseline predictive behavior while modestly improving probabilistic accuracy on many evaluation pairs.

4.3. Efficiency and Scaling Evaluation

We next test whether the change in recursion structure yields practical runtime gains on the solar benchmark while preserving predictive accuracy. Both methods use the same kernel family, spectral budget ($R = 1000$), dtype, device, and random seed. All timing experiments were run on a single NVIDIA GeForce RTX 4090 in double precision. We report wall-clock time for interdomain kernel construction as medians over 20 repeats after 5 warmup iterations, and we report test RMSE after fitting the variational parameters under fixed kernel hyperparameters. To complement wall-clock time, we additionally report profiler-estimated FLOP counts from PyTorch's `with_flops=True` profiler. These FLOP numbers are operator-level estimates rather than hardware peak throughput; for memory robustness they were collected with a reduced profiling budget $R = 512$ while keeping the same discretization-count and M sweeps. Because these are wall-clock GPU timings, small local non-monotonicities can arise from launch overhead, batching, and kernel scheduling; the FLOP profiles therefore provide the cleaner view of the asymptotic trend. Appendix F reports the corresponding summary tables.

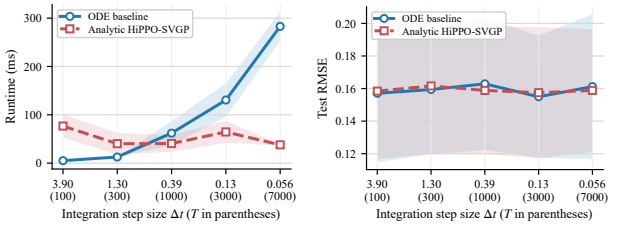
In the runtime sweeps, T denotes the number of HiPPO/ODE discretization steps over a fixed time interval, not the number of solar observations. The solar dataset contains only a few hundred samples, whereas the sweep uses $T \in \{100, 300, 1000, 3000, 7000\}$ to control the ODE integration resolution. The more faithful horizontal axis is therefore the integration step size

$$\Delta t = \frac{t_{\max} - t_{\min}}{T},$$

so increasing T corresponds to a finer temporal discretization rather than to a longer dataset. We therefore plot the runtime and RMSE curves against Δt, with the corresponding T values shown in parentheses on the tick labels.

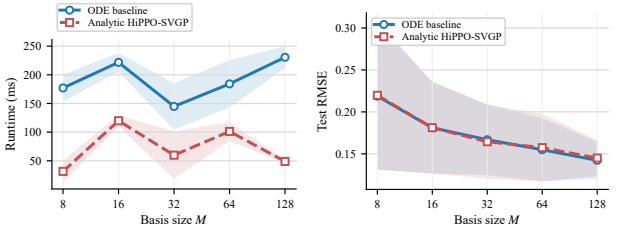
Figure 4 isolates the kernel-construction cost against Δt and compares the resulting test RMSE on the same sweep. The left panel is the direct runtime test of the theoretical claim in Section 3.2: as Δt decreases, the ODE baseline must traverse more temporal steps, whereas the analytic construction avoids that temporal recursion and therefore removes the dominant linear dependence on the discretization count. Quantitatively, the ODE kernel time rises from 5.1 ms at $T = 100$ to 282.9 ms at $T = 7000$, while the analytic kernel stays in a much flatter 37.8–76.8 ms band and is 7.5× faster at the finest discretization. The profiler-estimated kernel FLOP count shows the same trend more cleanly: the ODE path grows from 1.02 GFLOPs to 59.37 GFLOPs, whereas the analytic path remains essentially constant at ≈ 0.177 GFLOPs. The key evidence is therefore not that the analytic construction must be faster at every coarse discretization, but that the ODE cost grows with T whereas the analytic cost is nearly independent of T. The right panel checks the downstream predictive side of the same ablation. Here both methods remain in a narrow and overlapping RMSE band on solar (ODE: 0.155–0.163, analytic: 0.157–0.162), indicating that refining Δt mainly changes computational cost rather than predictive accuracy under the fixed-kernel evaluation used here.

Figure 5 performs the complementary sweep over basis size at fixed discretization, now using the finer setting $T = 3000$ (equivalently $\Delta t = 0.13$). The left panel again isolates kernel-construction time, while the right panel reports test RMSE at each M. Under this longer-horizon regime the runtime picture changes qualitatively: the analytic construction is faster than the ODE baseline across the entire tested M range. For kernel construction, the analytic medians range from 31.5 ms to 119.9 ms, whereas the ODE medians range from 144.8 ms to 230.6 ms. The same trend persists for one ELBO update step, where analytic ranges from 57.8 ms to



(a) Kernel construction *(b)* Test RMSE

Figure 4. Kernel runtime and predictive RMSE versus integration step size Δt at fixed $M = 64$ on the solar benchmark. Here $\Delta t = (t_{\max} - t_{\min})/T$, so the T values shown in parentheses on the x-axis are discretization counts rather than dataset size. Runtime lines show medians over 20 repeats after 5 warmup iterations; shaded bands denote one standard deviation. The RMSE panel shows that downstream predictive performance remains comparable across the tested discretizations even though the ODE construction cost grows sharply as Δt becomes smaller.



(a) Kernel construction *(b)* Test RMSE

Figure 5. Kernel runtime and predictive RMSE versus basis size M at fixed discretization $T = 3000$ on the solar benchmark. Runtime lines show medians over 20 repeats after 5 warmup iterations; shaded bands denote one standard deviation. At this finer discretization the analytic method is faster across the full tested M range while retaining essentially the same predictive accuracy as the ODE baseline.

125.4 ms and ODE ranges from 176.4 ms to 238.5 ms. We do not interpret the wall-clock curve as strictly monotone in M, since GPU launch overhead and batching can perturb the medians; instead, we use the FLOP estimates to read off the asymptotic trend. On the accuracy side, the two methods remain extremely close: both improve monotonically from about 0.22 at $M = 8$ to about 0.145 at $M = 128$, with only small fluctuations in between. The profiler-estimated FLOP counts support the same conclusion. At $M = 128$, for example, the analytic kernel requires only 0.236 GFLOPs versus 101.30 GFLOPs for ODE, so the wall-clock advantage at fixed $T = 3000$ is aligned with the arithmetic scaling trend rather than contradicting it.

5. Conclusion and Future Work

We presented an analytic replacement for ODE-based interdomain kernel construction in one-dimensional HiPPO-LegS OHSVGP. The central change is a shift from temporal ODE recursion to direct kernel evaluation together with recursion over polynomial order, implemented through spherical Bessel identities. Within this restricted setting, the resulting construction removes ODE discretization error from interdomain kernel construction and reduces its asymptotic dependence on the number of ODE integration steps needed to reach the current horizon, while maintaining comparable predictive performance on the evaluated benchmarks. The runtime study confirms that the expected reduction in discretization-step dependence appears in practice at fine Δt; at the finer fixed-discretization setting, the analytic construction is also faster across the tested basis sizes while maintaining comparable predictive accuracy. Important next steps are to study alternative basis functions and spectral approximations, to broaden the runtime evaluation across hardware and spectral budgets, to further optimize implementations of the Miller recursion and frequency batching for broader hardware evaluation, and to determine whether analogous analytic constructions are possible in higher-dimensional settings.

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A. RFF assembly and normalization

We spell out the normalization used in the RFF interdomain kernel assembly. For each sampled frequency ω_r, define the vectors

$$C(t;\omega_r) = \big(C_0(t;\omega_r), \dots, C_{M-1}(t;\omega_r)\big)^\top,$$

and

$$S(t;\omega_r) = \big(S_0(t;\omega_r), \dots, S_{M-1}(t;\omega_r)\big)^\top.$$

The interdomain feature matrix can be written compactly as

$$Z(t) = R^{-1/2}\big[C(t;\omega_1), \dots, C(t;\omega_R), S(t;\omega_1), \dots, S(t;\omega_R)\big] \in \mathbb{R}^{M\times 2R}.$$

Similarly, for data inputs $X = (x_1, \dots, x_N)$, define

$$c_r(X) = \big(\cos(\omega_r x_1), \dots, \cos(\omega_r x_N)\big)^\top,$$

and

$$s_r(X) = \big(\sin(\omega_r x_1), \dots, \sin(\omega_r x_N)\big)^\top,$$

so that

$$Z_f(X) = R^{-1/2}\big[c_1(X), \dots, c_R(X), s_1(X), \dots, s_R(X)\big] \in \mathbb{R}^{N\times 2R}.$$

With this convention,

$$Z(t)Z(t)^\top = \frac{1}{R}\sum_{r=1}^{R}\left[C(t;\omega_r)C(t;\omega_r)^\top + S(t;\omega_r)S(t;\omega_r)^\top\right].$$

Elementwise, this gives

$$[Z(t)Z(t)^\top]_{\ell m} = \frac{1}{R}\sum_{r=1}^{R}\left[C_\ell(t;\omega_r)C_m(t;\omega_r) + S_\ell(t;\omega_r)S_m(t;\omega_r)\right],$$

which is the Monte Carlo approximation to the Bochner-factorized inducing covariance $K_{uu}(t)$ for stationary kernels (Rudin, 1990; Rahimi and Recht, 2007). Likewise,

$$[Z_f(X)Z(t)^\top]_{n\ell} = \frac{1}{R}\sum_{r=1}^{R}\left[\cos(\omega_r x_n)C_\ell(t;\omega_r) + \sin(\omega_r x_n)S_\ell(t;\omega_r)\right],$$

which gives the corresponding RFF approximation to $K_{fu}(t)$. Therefore the analytic construction changes only the evaluation of the overlap vectors $C(t;\omega_r)$ and $S(t;\omega_r)$; the RFF kernel assembly remains the same as in the OHSVGP construction (Chen et al., 2025).

B. Numerical details of backward Miller recursion

The analytic construction requires evaluating $\{j_\ell(\kappa)\}_{\ell=0}^{M-1}$ for each sampled frequency. Direct upward recurrence can be unstable when the order exceeds the argument, because the recurrence may amplify the irregular solution. We therefore use backward Miller recursion (Gautschi, 1967; DLMF).

For a target maximum order $M-1$, we choose a truncation order L larger than the maximum order of interest, with a safety margin chosen sufficiently large for the range of κ encountered in the experiment. In practice, L can be increased until the normalized values of $j_0(\kappa), \dots, j_{M-1}(\kappa)$ are insensitive to further increases. Starting from an arbitrary nonzero seed pair

$$\tilde{j}_L = 1, \qquad \tilde{j}_{L+1} = 0,$$

we run the three-term recurrence backward,

$$\tilde{j}_{\ell-1}(\kappa) = \frac{2\ell+1}{\kappa}\tilde{j}_\ell(\kappa) - \tilde{j}_{\ell+1}(\kappa), \qquad \ell = L, L-1, \dots, 1.$$

The resulting sequence is correct only up to a common scale. We fix this scale using the known value

$$j_0(\kappa) = \frac{\sin \kappa}{\kappa},$$

and set

$$j_\ell(\kappa) = \tilde{j}_\ell(\kappa) \frac{\sin \kappa}{\kappa \, \tilde{j}_0(\kappa)}, \qquad \ell = 0, \ldots, M-1.$$

For $|\kappa|$ near zero, the normalization above is replaced by the small-argument expansion

$$j_\ell(\kappa) = \frac{\kappa^\ell}{(2\ell+1)!!} + \mathcal{O}(\kappa^{\ell+2}),$$

with the limiting value $j_0(0) = 1$. This avoids division by zero and stabilizes the low-frequency case. The backward pass requires $\mathcal{O}(L)$ arithmetic operations per frequency. For the bounded range of κ encountered in our experiments, we use a fixed safety margin so that L scales linearly with M, yielding $\mathcal{O}(M)$ work per frequency.

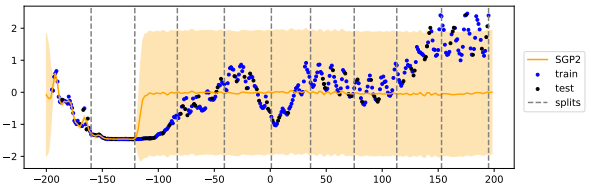
C. Additional experimental details

All matched analytic-versus-ODE comparisons use the same data split, likelihood, kernel family, initialization, inducing budget, spectral budget, dtype, device, and random seed unless otherwise stated. The analytic and ODE variants differ only in the construction of the HiPPO-LegS interdomain feature matrix $Z(t)$. Predictive metrics are computed after fitting the same variational objective under the matched setup. Runtime measurements isolate interdomain kernel construction unless explicitly reported as full ELBO update time.

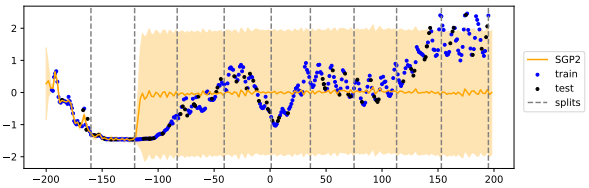
D. Additional Solar Results

Table 3. Task-wise solar irradiance comparison between the ODE-based and analytic variants. The main Solar runs use basis size $M = 100$ and ODE step size $\Delta t = 2.45$. Negative deltas indicate an improvement of the analytic method over the ODE baseline.

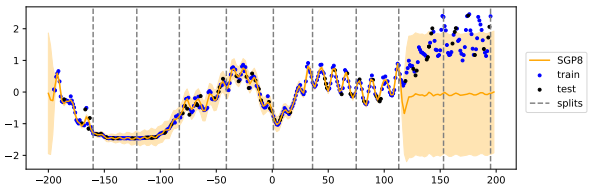
Task	ODE RMSE	Analytic RMSE	Δ RMSE	ODE NLL	Analytic NLL	Δ NLL
1	0.1992	0.1802	-0.0190	8.0143	5.4882	-2.5261
2	0.1346	0.1146	-0.0201	4.5787	1.4713	-3.1074
3	0.1060	0.0911	-0.0149	1.9400	0.1451	-1.7949
4	0.1643	0.1621	-0.0021	7.8561	6.4982	-1.3578
5	0.1534	0.1509	-0.0025	6.5244	5.4849	-1.0395
6	0.1412	0.1457	0.0045	5.3541	2.5755	-2.7786
7	0.1298	0.1449	0.0152	1.3365	0.8754	-0.4611
8	0.1173	0.1467	0.0294	0.8595	1.5299	0.6704
9	0.1429	0.1296	-0.0133	1.5256	0.0976	-1.4281
10	0.1598	0.1521	-0.0077	2.3635	0.6493	-1.7143
Mean	0.1448	0.1418	-0.0030	4.0353	2.4815	-1.5537



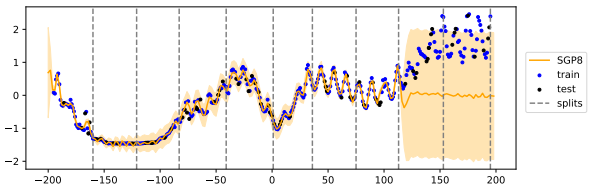
(a) Task 2, ODE



(b) Task 2, Analytic

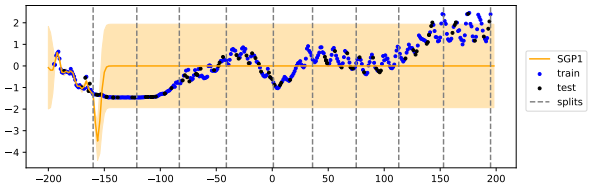


(c) Task 8, ODE

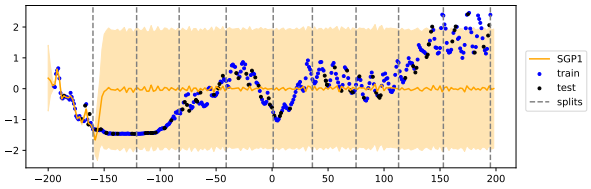


(d) Task 8, Analytic

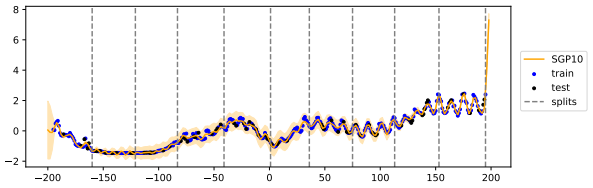
Figure 6. Representative solar posterior comparisons. Task 2 shows the strongest improvement of the analytic construction in both RMSE and NLL, while Task 8 provides the main counterexample where the ODE construction performs better.



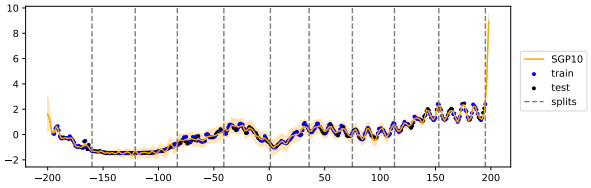
(a) Task 1, ODE



(b) Task 1, Analytic



(c) Task 10, ODE



(d) Task 10, Analytic

Figure 7. Early- and late-stream posterior comparisons on solar irradiance. The analytic construction remains competitive at the beginning of the stream and retains an advantage in the later stage, consistent with the improved aggregate NLL.

E. Additional COVID Results

Table 4. Task-wise COVID-19 mortality comparison between the ODE-based and analytic variants. The main COVID runs use basis size $M = 15$ and ODE step size $\Delta t = 2$. Negative deltas indicate an improvement of the analytic method over the ODE baseline.

Eval	After	ODE NLPD	Analytic NLPD	Δ NLPD	ODE ECE	Analytic ECE	Δ ECE
1	1	2.9599	2.9636	0.0037	0.1500	0.1463	-0.0037
1	2	2.9783	2.9489	-0.0294	0.1278	0.1278	0.0000
2	2	3.7922	3.7910	-0.0011	0.0800	0.0717	-0.0083
1	3	2.9806	2.9527	-0.0279	0.1352	0.1352	0.0000
2	3	3.6662	3.6625	-0.0037	0.0767	0.0767	0.0000
3	3	3.8487	3.8628	0.0141	0.1183	0.1117	-0.0067
1	4	3.0697	2.9829	-0.0869	0.1907	0.1426	-0.0481
2	4	3.6496	3.6491	-0.0006	0.0767	0.0867	0.0100
3	4	3.8816	3.8965	0.0149	0.1283	0.1200	-0.0083
4	4	3.8058	3.8005	-0.0053	0.0517	0.0433	-0.0083
1	5	3.0870	3.0674	-0.0195	0.1537	0.1426	-0.0111
2	5	3.6310	3.6207	-0.0102	0.0867	0.1183	0.0317
3	5	3.8031	3.7754	-0.0277	0.0750	0.0767	0.0017
4	5	3.9312	3.9027	-0.0285	0.0533	0.0583	0.0050
5	5	4.1404	4.2054	0.0650	0.1375	0.1750	0.0375
Overall mean		3.5484	3.5388	-0.0095	0.1094	0.1089	-0.0006
Final mean		3.7185	3.7143	-0.0042	0.1012	0.1142	0.0129

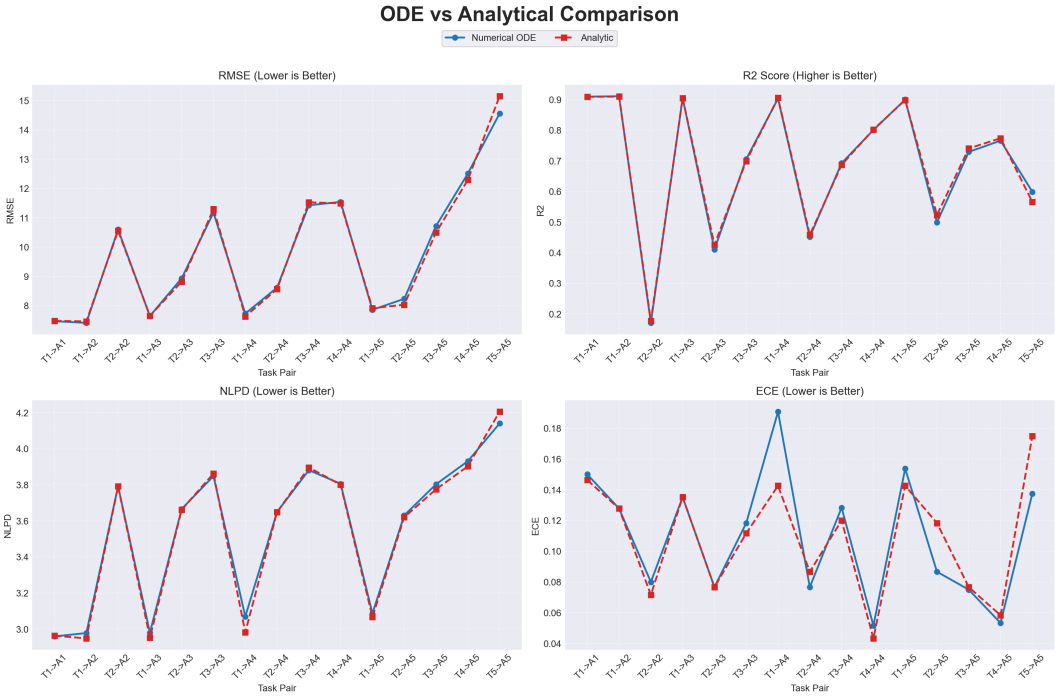
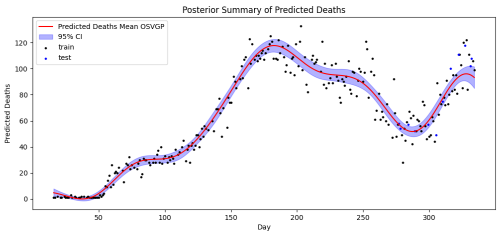
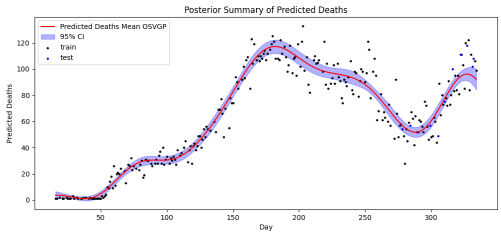


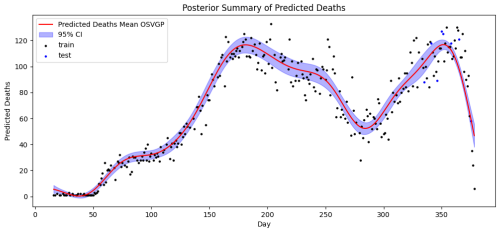
Figure 8. Task-pair comparison across the COVID-19 stream. The analytic construction closely tracks the ODE baseline in RMSE and R^2, improves NLPD on most evaluation pairs, and yields its clearest calibration gain around Task 4, while the final Eval 5 after Task 5 pair remains the main counterexample.



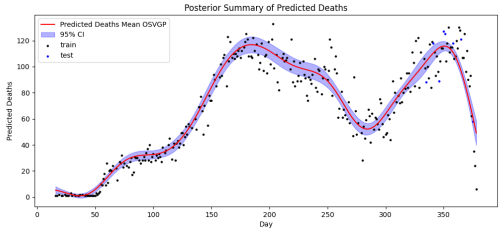
(a) Task 4, ODE



(b) Task 4, Analytic



(c) Task 5, ODE



(d) Task 5, Analytic

Figure 9. Representative COVID-19 posterior comparisons in a two-column layout. Task 4 illustrates a case where the analytic construction slightly improves predictive density while largely preserving the posterior shape of the ODE baseline, whereas Task 5 shows the main final-stage counterexample where the analytic variant underperforms.

F. Additional Runtime Results

Because these are wall-clock medians on GPU kernels with non-negligible launch overhead, small local inversions between kernel-only and full-update timings should not be over-interpreted.

Table 5. Solar kernel/runtime summary for the discretization sweep at fixed $M = 64$ and $R = 1000$. Here T is the number of HiPPO/ODE integration steps and $\Delta t = (t_{\max} - t_{\min})/T$. Times are median wall-clock milliseconds over 20 repeats after 5 warmup iterations.

T	Analytic kernel	ODE kernel	Analytic full update	ODE full update
100	76.8	5.1	53.1	11.1
300	40.2	12.7	63.2	19.0
1000	40.6	62.1	49.7	80.1
3000	64.5	130.6	77.1	133.0
7000	37.8	282.9	44.6	289.5

Table 6. Solar kernel/runtime summary for the basis-size sweep at fixed discretization $T = 3000$ and $R = 1000$. Times are median wall-clock milliseconds over 20 repeats after 5 warmup iterations.

M	Analytic kernel	ODE kernel	Analytic full update	ODE full update
8	31.5	177.1	72.3	176.4
16	119.9	221.7	125.4	235.7
32	59.9	144.8	57.8	225.7
64	101.2	184.2	108.0	194.4
128	48.9	230.6	60.7	238.5

Table 7. Profiler-estimated FLOP summary for the discretization sweep at fixed $M = 64$. Values are reported in GFLOPs per call. These profiler totals use the same T sweep but a reduced profiling budget $R = 512$ for memory robustness.

T	Analytic kernel	ODE kernel	Analytic full update	ODE full update
100	0.177	1.022	0.253	1.098
300	0.177	2.713	0.253	2.789
1000	0.177	8.633	0.253	8.709
3000	0.177	25.545	0.253	25.621
7000	0.177	59.370	0.253	59.446

Table 8. Profiler-estimated FLOP summary for the basis-size sweep at fixed discretization. Values are reported in GFLOPs per call. These profiler totals use the same M sweep but a reduced profiling budget $R = 512$ for memory robustness.

M	Analytic kernel	ODE kernel	Analytic full update	ODE full update
8	0.139	0.563	0.178	0.601
16	0.144	1.772	0.187	1.814
32	0.154	6.549	0.206	6.602
64	0.177	25.545	0.253	25.621
128	0.236	101.298	0.388	101.450